

6(a)	2.0 cm	B1
6(b)	At 16 (cm) from A the electric fields are equal or $E_A = E_B$	B1
	$E = Q / 4\pi\epsilon_0 r^2$	C1
	$Q_A / (4\pi\epsilon_0 r_A^2) = Q_B / (4\pi\epsilon_0 r_B^2)$	
	$3.6 \times 10^{-9} / 0.16^2 = Q_B / 0.08^2$	
	$Q_B = \mathbf{9.0 \times 10^{-10} C}$	A1
6(c)(i)	$V = Q / 4\pi\epsilon_0 r_A$	C1
	$V = 3.6 \times 10^{-9} / (4 \times \pi \times 8.85 \times 10^{-12} \times 0.020)$	
	$V = \mathbf{1600 V}$	A1
6(c)(ii)	$C = Q / V$	C1
	$= 3.6 \times 10^{-9} / 1600$	
	$= \mathbf{2.3 \times 10^{-12} F}$	A1